

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

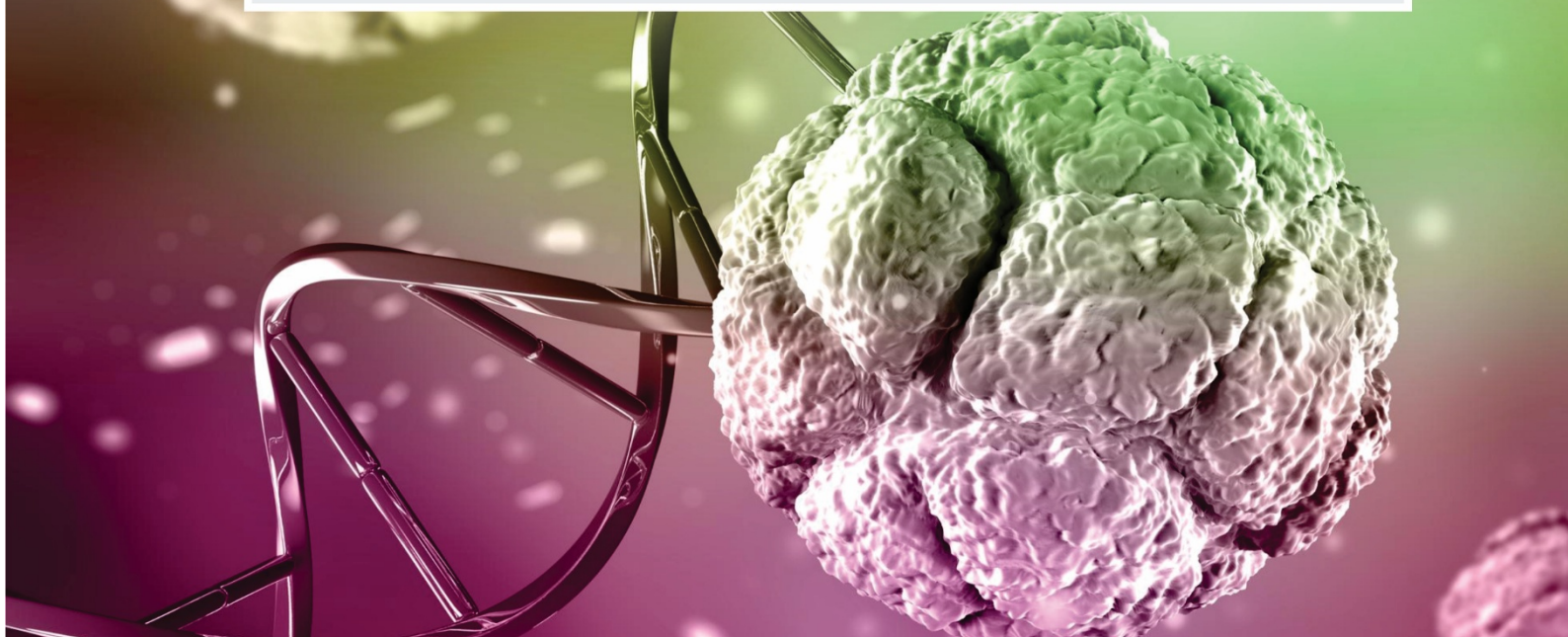
BIOLOGY

Topic wise
Notes

LECTURE

21

Human Endocrine System



Human Endocrine System

Human Endocrine System

- The human endocrine system is a network of glands that produce hormones to regulate body functions like growth, metabolism, and reproduction.
- Key glands include the hypothalamus, pituitary, thyroid, parathyroids, adrenal glands, pancreas, ovaries, and testes.
- The branch of science which deals with study of endocrine glands is known as endocrinology.
- Thomas Adison is known as Father of Endocrinology.
- Hormone term was given by starling.
- First Hormone discovered – secretion.
- Hormone can be made up of amino acid or peptide or steroids (lipids)
- Hormones are known as chemical messengers.

Gland System

A gland is a specialized organ or group of cells that produces and secretes substances needed for the body's functioning.

Types of Gland

1. Exocrine Glands

- Secrete substances through ducts (tubes) to a specific location
- Secretions are not hormones

Examples:

- Salivary glands saliva
- Sweat glands sweat
- Sebaceous glands - oil

2. Endocrine Glands

- Ductless glands
- Release hormones directly into the bloodstream

Examples:

- Pituitary gland
- Thyroid gland
- Adrenal gland

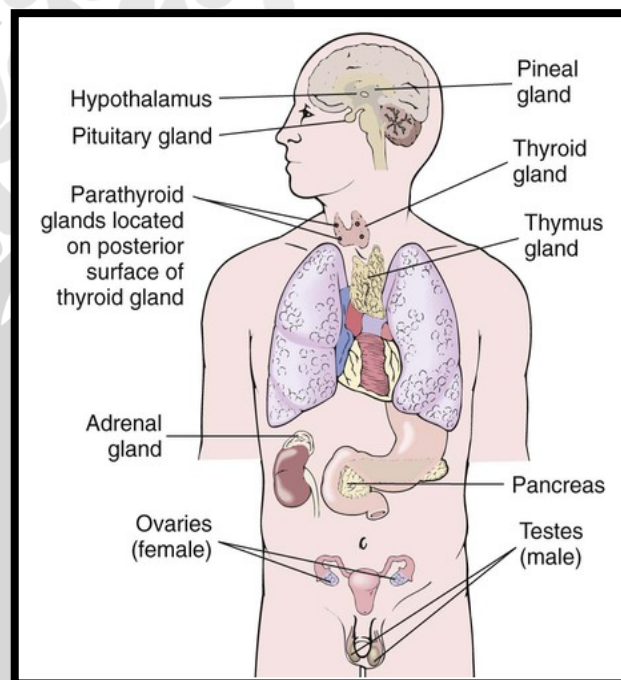
3. Mixed Glands (Heterocrine Glands)

- Have both endocrine and exocrine functions

Example:

- Pancreas
- Endocrine insulin, glucagon
- Exocrine digestive enzymes

Glands		
Endocrine glands	Exocrine glands	Mixed Glands
Ductless gland Secretum direct Poured into blood	Gland with ducts. Secretions are transported through their duct only at target organ.	Having both types of secretory cells. It means they can have ducts & secrete without duct also.



- Each gland has specific functions, such as pituitary regulating other glands, the thyroid controlling metabolism, and the pancreas managing blood sugar through insulin and glucagon.

Gland	Location	Hormones produced	Primary Functions
Hypothalamus	Base of the brain	Gonadotropin-releasing hormone (GnRH), corticotropin-releasing hormone (CRH) etc.	Links the nervous and endocrine system; control the pituitary gland.
Pituitary	Base of the brain	Growth Hormone, TSH, ACTH, FSH, LH, prolactin	'master gland' regulates other endocrine glands and many body function.

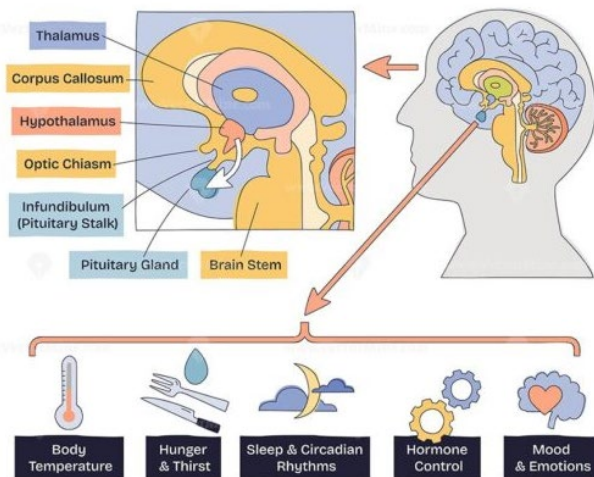




Hypothalamus

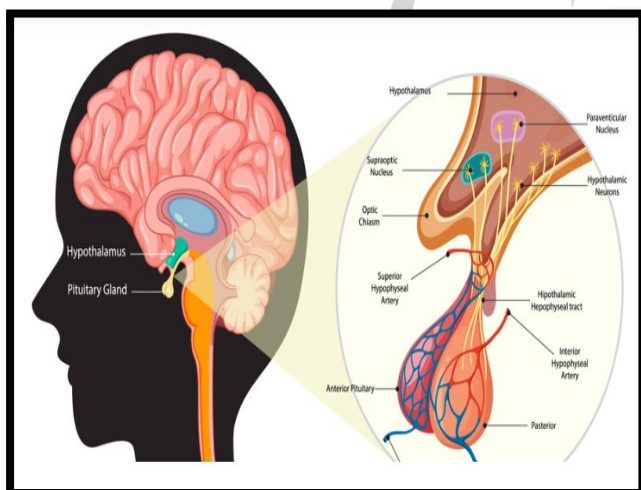
The hypothalamus is a small but vital region of the brain located below the thalamus and above the brainstem, responsible for maintaining body homeostasis.

The Hypothalamus Brain's Control Center



NUCLEUS	STIMULATION OF	LESION OF
Suprachiasmatic nucleus	Adjusts the circadian clock phase	Abolishes circadian rhythms
Supraoptic or paraventricular nuclei	Increased blood volume, blood pressure, and metabolism	Diabetes insipidus
Lateral hypothalamic nucleus	Increased feeding	Decreased feeding
Ventromedial nucleus	Decreased feeding	Increased feeding
Dorsomedial nucleus	Sham rage	Decreased aggression and decreased feeding
Mammillary body	?	Short-term memory is not processed into long-term memory

Pituitary Gland



Hypothalamic factor	Hormone	Structure	Target cell	Feedback control Hormone or compound
GHRH (s)	GH	Protein	Bone, soft tissue	GH release-inhibiting hormone
Dopamine	PRL	Protein	Breast	Dopamine
CRH	ACTH	Protein	Adrenal cortex	Cortisol
GnRH (s)	LH	Glycoprotein	Gonads	Gonadal steroid
GnRH (s)	FSH	Glycoprotein	Gonads	Gonadal steroid and inhibin
TRH (s)	TSH	Glycoprotein	Thyroid gland	Free T4, free T3

Posterior Pituitary

Oxytocin hormone

- Love Hormone
- Parental Care
- Labour pain causing Hormone
- After parturition, it stimulates milk ejection

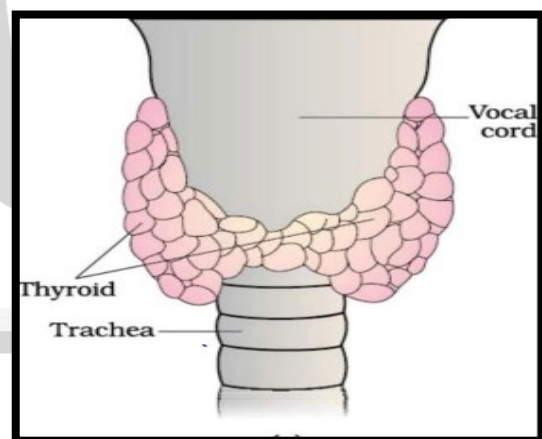
Antidiuretic hormone/ Vasopressin

- Concentration of urine
- Hypo Secretion
- Diabetes insipidus

Human Endocrine System

Gland	Location	Hormones Produced	Primary Functions
Thyroid	Neck	Thyroxine (T_4), Triiodothyronine (T_3)	Regulates metabolism, body temperature, and heart rate.
Parathyroid	Neck, on the back of the thyroid	Parathyroid hormone (PTH)	Regulates calcium and phosphorus levels in the blood.

Thyroid Gland



- Present at the ventral surface of larynx (voice box).
- Bilobed structure.
- Largest Endocrine Gland.
- Two types of secretions:
 - Thyroxine
 - Thyrocalcitonin/ Calcitonin



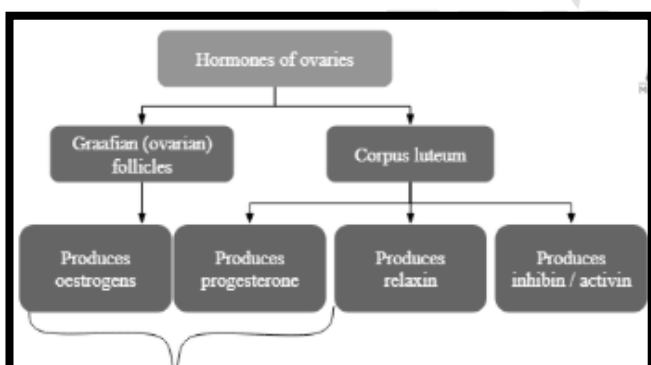
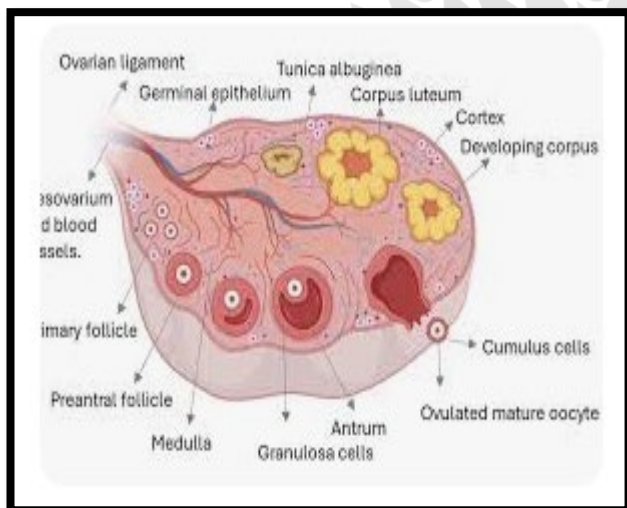
Thyroxin — Physical and mental growth development

Function

1. Increase BMR (Basic metabolic rate)
2. Increases heart Rate
3. Regulate neural impulse transmission
 - **Fat** → Fatty acid glycerol
 - **Carb** → Glucose
 - **Protein** → Amino Acid

OVARY

The ovaries are paired female reproductive glands, roughly almond-shaped, located in the pelvis on both sides of the uterus.



Gland	Location	Hormones Produced	Primary Function
Pineal	Brain	Melatonin	Regulates the sleep-wake cycle and circadian rhythm
Thymus	Chest	Thymosin	Develops the immune system in youth.

Hyposecretion

Hyposecretion is the abnormally low production and release of hormones by an endocrine gland, causing deficiency disorders.

Sl. no	Effects of thyroid hormones	Hypersecretion	Hyposecretion
1.	Basal metabolic rate	BMR can be increased to about 60-100% above normal. The person is hyperactive.	BMR decreases to about half the normal rate. The person becomes sluggish.
2.	Carbohydrate metabolism	Increased uptake of glucose by cells. Increased gluconeogenesis. Increased breakdown of glucose.	Decrease in uptake of glucose by cells. Decreased gluconeogenesis. Decreased breakdown of glucose.
3.	Fat metabolism	Fat reserves are depleted, increased oxidation of the free fatty acids. Increases the excretion of cholesterol in the bile.	Fat metabolism is slowed down.

- Pretension → (child)
- Myxedema (Adult) (weight gain obesity heart rate)
- Simple Goitre (Mainly due to iodine deficiency in diet.)
- Hashimoto disease (Autoimmune disorder)

Hypersecretion

- Exophthalmic goiter or Grave's disorder

GLAND/ORGAN	HORMONE(S) SECRETED/RELEASED	ACTIONS	HYPERSECRETION	HYPOSECRETION
PITUITARY ANTERIOR LOBE (adenophysis)	Adrenocorticotropic hormone - stimulated by CRH from hypothalamus	Stimulates adrenal cortex to produce cortisol, corticosteroids, and androgens; can stimulate melanocytes	• Pituitary ACTH hypersecretion or (Cushing Disease) • Congenital adrenal hyperplasia	• Adrenocorticotropic hormone deficiency (ACTH deficiency) • Cushing's syndrome
	Growth Hormone (GH or HGH), somatotrophic hormone - stimulated by GRH, GHRH from hypothalamus	Controls bone and tissue growth and regulates metabolism (influences secretion of insulin-like growth factor I from liver)	• Gigantism (preadult)-abnormal tall (~ 8ft) • Acromegaly (mature adult)	• Dwarfism (pre-adult) • Adults with growth hormone deficiency feel tired to a level that impairs their wellbeing
	Thyroid-stimulating hormone (TSH) - stimulated by TRH from hypothalamus	Regulates thyroid hormone (via secretion of thyroxine [T ₄] and triiodothyronine [T ₃])	• Hyperthyroidism • Grave's disease	• Hypothyroidism • Myxedema - Physical and mental sluggishness (not retardation), puffiness of face, fatigue, poor muscle tone, low body temperature, obesity, dry skin
	Follicle-stimulating hormone (FSH) - stimulated by GnRH	Stimulates growth and secretion of eggs in ovaries (female) and sperm in testes (male)	• Understimulation of adrenal cortex hormones • Kallmann syndrome	• Isolated FSH deficiency • bPantial follicle stimulating hormone deficiency in men can cause delayed puberty and limited sperm production (oligozoospermia), but fathering a child may still be possible.
	Luteinizing hormone (in females)	Helps control ovulation and menstruation; important in sustaining pregnancy	• Causes infertility and miscarriage • Multiple births (for women)	• lack of sexual development and sterility • Hypogonadism